# On the Decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ Concept Satisfiability: Split-Forest Models and a Machine-Checked Reduction to One Keystone

Michael Wessel
(with AI assistants: Claude / Anthropic, GPT-5.5 / OpenAI)

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**Abstract**

Whether concept satisfiability in $\mathcal{ALCI}_{\mathrm{RCC5}}$—the description logic $\mathcal{ALCI}$ with the RCC5 base relations as roles under *abstract composition-table* semantics—is decidable has been open since Wessel's 2002/2003 work. This paper reports a sustained, adversarially-reviewed, partly machine-checked attack on the problem, conducted by AI assistants under human direction. We do not settle the question. What we establish is: (i) a *conditional decidability theorem*—if the "live width" of models is bounded by a computable function of the concept (property F6), then satisfiability is decidable—whose soundness direction is *machine-checked in Lean 4*; (ii) that the entire remaining difficulty compresses into that single property F6, which we show is the *same fact* that blocks a proof of undecidability; (iii) a certified fragment of the local algebra (an RCC5 normal form) and an unconditionally decidable fragment (the $\forall$PO-free concepts); and (iv) an honest map of the routes tried, the dead ends, and the missing hard pieces. We also report, candidly, on the methodology—adversarial "cold review," machine verification, and a human-in-the-loop workflow—and on what frontier AI could and could not do.

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# 1 Introduction

## 1.1 AI disclaimer and authorship

The mathematical content of this paper was produced by AI assistants (Anthropic's Claude models and OpenAI's GPT-5.5), prompted, steered, and reviewed by the human author (Michael Wessel), who posed the problem, supplied the central intuitions and directions, orchestrated the review–repair cycles, and takes responsibility for the framing. **The results are unverified by human domain experts and should be read as AI-generated conjectures supported by machine checking and computational evidence—not as established theorems**, with the explicit exceptions of the parts stated to be kernel-checked in Lean 4 (the soundness pipeline, the abstraction's faithfulness, and the RCC5 normal form), which are machine-verified but still human-unreviewed. We are deliberate about this labelling throughout: the standing one-line status is *strongly supported, not certified*. We regard the AI systems as having made contributions of a kind that, in a human collaborator, would warrant co-authorship, and we discuss this in §8.

## 1.2 Why the problem is hard, in one page

RCC5 relates two regions by exactly one of five relations: equal (EQ), proper part (PP), proper superpart (PPI), partial overlap (PO), or discreteness/disjointness (DR). Under *abstract composition-table* semantics a model is not a set of actual regions but a set with a total labelling of ordered pairs by these relations, required only to be converse-closed and *composition-closed*: if $x\,R\,y$ and $y\,S\,z$ then $x\,T\,z$ for some $T$ the composition table allows.

The five relations split into two axes that behave completely differently.

- **The vertical axis** (PP, PPI: nesting) is *tame*. Composition can *force* vertical structure—the only four "determining" compositions (those returning a single relation) all run through PP or PPI—but an unbounded vertical chain folds into a finite loop, exactly as in ordinary modal logic. Infinite depth is not the problem.
- **The horizontal axis** (DR, PO: side-by-side) is *wild*. Composition can never *determine* a horizontal relation (PO is never a forced value), yet it can force two occurrences to be *distinct* (a shared horizontal anchor excludes EQ). So horizontal "crowds" of pairwise-distinct occurrences are the danger.

A finite decision procedure must record, at each interface of a model, the genuine branching it cannot reconstruct—the *live width*. Many forced edges are *shadows* (implied by composition from edges already present) and cost nothing; live width counts only the rest. The whole decidability question reduces to one property:

**(F6)** Is the live width of models bounded by a computable function of the input concept?

If yes, the certificate space is finite and satisfiability is decidable. And here is the knife's edge: F6 is *the same fact* that blocks proving the problem *undecidable*. To force undecidability one would encode a rigid coordinate grid, which requires forcing a horizontal relation to a single value—exactly what the composition table cannot do (the "coincidence obstruction" Wessel identified in 2002/2003). Turn that looseness into a bound and one gets decidability; circumvent it and force a grid and one gets undecidability. The two open directions are one question seen from two sides, which is why neither has been settled in over twenty years.

### 1.3 Scope, contributions, and structure

This is both a technical report and a discussion piece. Its contributions are:

1. A *conditional decidability theorem* (F6 $\Rightarrow$ decidable) whose soundness half is machine-checked in Lean 4 (§6.5), and the reduction of the whole problem to F6.
2. The identification of F6 as the shared keystone of the decidability and undecidability directions (§1.2, §9).
3. A certified RCC5 *normal form* and an unconditionally decidable $\forall\mathsf{PO}$-*free* fragment (§6.2, §6.6).
4. An honest, technique-organized account of every route tried, the counterexamples that killed the wrong ideas, and the lessons (§6), so that others need not repeat them.
5. A candid methodology report on adversarial AI-assisted mathematics (§8).

The paper is organized as follows. §2 states, in largely non-technical terms, where the problem stands after twenty-five years. §4 gives the history ($\mathcal{ALCI}_{\mathrm{RCC}}$ from Cohn 1993 through Wessel 2002/2003 and Lutz–Wolter 2006) and carefully distinguishes our object from the neighbouring logics that *are* known undecidable. §5 explains why the standard undecidability reductions fail. §6—the core of the paper—surveys the approaches *by technique*: naive tableaux and blocking, finite-model and reduction attempts, the split-forest idea, the automata route, the Lean line culminating in F6, and the regular-cover pivot. §7 describes the working reasoners and a GIS example. §8 reports on the AI methodology. §9 concludes with what was learned, what would move the needle, an honest estimate of the chances of closing F6, and the project's actual wins.

## 2 Current status: where we stand after twenty-five years

*Decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept satisfiability remains open* (the closely related $\mathcal{ALCI}_{\mathrm{RCC8}}$ likewise, but it is not our focus here). A current-literature check (July 2026) found no published resolution; the closest neighbours are decided in the other direction and are discussed in §4. What the present work adds is not an answer but a sharp reduction and a machine-checked surround.

**The positive result—conditional decidability.** *If* the live width of models is bounded by a computable function of the concept (property F6), *then* concept satisfiability is decidable; and the **soundness half of that reduction is machine-checked in Lean 4** (a valid finite certificate provably unfolds to a genuine RCC5 model of the concept, and the finite type abstraction is faithful). What is open is the antecedent F6 itself—equivalently the completeness direction, that every satisfiable concept has a *bounded* finite certificate.

**What is certified** (machine-checked, Lean 4, zero `sorry`):

- the *soundness* pipeline: a valid finite certificate unfolds to a genuine RCC5 model of the concept;
- the *faithfulness* of the finite (Hintikka) abstraction: satisfiability and abstraction-realizability coincide;
- the *RCC5 normal form* (forward): every strong-$\mathsf{EQ}$ RCC5 network is an ordered-disjoint structure—a strict partial order for $\mathsf{PP}$ with a downward-closed disjointness for $\mathsf{DR}$, $\mathsf{PO}$ residual.

**What is strongly supported but not certified:** F6, the single open keystone—now sharpened to a precise combinatorial dichotomy (does some concept force unbounded "identity-selector minors"?), with the auxiliary uniformization property W2′ shown to *fold into* it.

**Unconditional by-products (independent of F6):**

- the $\forall\mathsf{PO}$-free fragment is *decidable*—concepts with no $\forall\mathsf{PO}$ subformula, still retaining $\exists\mathsf{PO}$, $\forall\mathsf{DR}$, $\exists\mathsf{DR}$, and all vertical modalities, so a genuinely expressive spatial fragment;
- the certified RCC5 normal form above;
- working reasoners (a cover-tree tableau) that reproduce, 23 years later, the GIS taxonomy of Wessel's original report;
- the standard undecidability routes shown blocked, with the coincidence obstruction identified as the shared fact.

**The honest bottom line.** After roughly thirty repair rounds and fifteen adversarial reviews the project has reached a genuine *local optimum*: the local algebra is exhausted (and in one case certified), the soundness side is machine-checked, and the whole remaining difficulty is one well-posed lemma. We offer the work not as a resolution but as a two-decade open problem *reduced, honestly and checkably, to its irreducible core.*

# 3 Preliminaries

We fix the object logic precisely; readers familiar with $\mathcal{ALCI}_{\mathrm{RCC}}$ may skim to Definition 3.

**Syntax.** $\mathcal{ALCI}_{\mathrm{RCC5}}$ concepts are built from atomic concepts $A$ and the five RCC5 relations, used as roles, by

$$C ::= A \mid \neg A \mid \top \mid \bot \mid C \sqcap C \mid C \sqcup C \mid \exists R.C \mid \forall R.C, \qquad R \in \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}.$$

We work in *negation normal form* (NNF: negation only on atoms) and *absorb inverse roles* through the RCC5 converse—$\exists\mathsf{PP}^{-}.C = \exists\mathsf{PPI}.C$, $\exists\mathsf{PO}^{-}.C = \exists\mathsf{PO}.C$, and so on—so no separate inverse constructor is needed. Let $\mathsf{cl}(C_0)$ be the Fischer–Ladner closure of the input concept and $m = |\mathsf{cl}(C_0)|$.

**Semantics: abstract composition tables, strong $\mathsf{EQ}$.** An interpretation is a pair $\mathcal{I} = (\Delta, \rho)$ with $\Delta \neq \emptyset$ and a *total* labelling $\rho : \Delta \times \Delta \to \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$ satisfying: (i) *strong* $\mathsf{EQ}$: $\rho(x, y) = \mathsf{EQ}$ iff $x = y$ (so $\mathsf{EQ}$ is identity, and between distinct elements only $\{\mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$ occur); (ii) *converse-closed*: $\rho(y, x) = \mathrm{conv}(\rho(x, y))$, where $\mathsf{PP}^{-1} = \mathsf{PPI}$, $\mathsf{PO}^{-1} = \mathsf{PO}$, $\mathsf{DR}^{-1} = \mathsf{DR}$; and (iii) *composition-closed*: $\rho(x, z) \in \mathsf{comp}(\rho(x, y), \rho(y, z))$ for all $x, y, z$, over the RCC5 composition table (Table 1). Concept extensions are standard, reading a role $R$ as the pair set $\{(x, y) : \rho(x, y) = R\}$; the concept $C_0$ is *satisfiable* if $x \in C_0^{\mathcal{I}}$ for some $\mathcal{I}$ and some $x \in \Delta$. Thus a model is a complete graph carrying a composition-closed RCC5 edge-labelling; *models may be infinite*. This is precisely the *abstract* (composition-table) semantics—not a semantics of actual regions in a topological space; weak-$\mathsf{EQ}$ semantics reduces to strong via TBox internalization, so we fix strong $\mathsf{EQ}$ throughout. The decision problem is: given $C_0$, is it satisfiable?

**The sister logic $\mathcal{ALCI}_{\mathrm{RCC8}}$.** $\mathcal{ALCI}_{\mathrm{RCC8}}$ is defined identically, over the eight RCC8 relations $\{\mathsf{DC}, \mathsf{EC}, \mathsf{PO}, \mathsf{EQ}, \mathsf{TPP}, \mathsf{NTPP}, \mathsf{TPPI}, \mathsf{NTPPI}\}$, which *refine* RCC5 by splitting $\mathsf{DR}$ into $\mathsf{DC}$ (disconnected) and $\mathsf{EC}$ (externally connected, i.e. touching at the boundary), and $\mathsf{PP}, \mathsf{PPI}$ into tangential and non-tangential parts ($\mathsf{TPP}, \mathsf{NTPP}$ and $\mathsf{TPPI}, \mathsf{NTPPI}$); $\mathsf{PO}$ and $\mathsf{EQ}$ are unchanged. Its abstract composition-table semantics is the exact analogue over the RCC8 composition table, and RCC8 likewise has the patchwork property. $\mathcal{ALCI}_{\mathrm{RCC5}}$ is our focus; we invoke $\mathcal{ALCI}_{\mathrm{RCC8}}$ only where it matters—its ExpTime-hardness (§4) and the domino attempts (§5), which turn on the extra $\mathsf{TPP}/\mathsf{NTPP}$ distinction.

**The two axes, made precise.** Call $\mathsf{PP}, \mathsf{PPI}$ *vertical* (nesting) and $\mathsf{DR}, \mathsf{PO}$ *horizontal* (side-by-side). The composition table makes the axes behave oppositely, and this asymmetry is the root of the whole difficulty.

**Proposition 1** (Determination is vertical)**.** *Of the sixteen compositions of proper (non-*$\mathsf{EQ}$*) relations, exactly four are* determining *(return a singleton):*

$$\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}, \quad \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}, \quad \mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}, \quad \mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}.$$

*Every one has a vertical (*$\mathsf{PP}$ *or* $\mathsf{PPI}$*) argument, and* $\mathsf{PO}$ *is never the value of any determining composition. (Exhaustive over Table 1; machine-verified.)*

So composition can *force* an edge only through the vertical axis; it can force two occurrences to be *distinct* (a shared horizontal anchor excludes $\mathsf{EQ}$) but never to a single horizontal value. Vertical recurrence, moreover, always folds:

**Lemma 2** (No horizontal recurrence)**.** *Form the composition-path subset automaton whose states are the reachable relation sets, with* $\mathrm{Rel}(\varepsilon) = \{\mathsf{EQ}\}$ *and* $\mathrm{Rel}(w\,a) = \bigcup_{r \in \mathrm{Rel}(w)} \mathsf{comp}(r, a)$*, restricted* *to states not containing* EQ. *Every cycle in this automaton uses only vertical labels; no cycle uses a horizontal label. Hence any support recurrence that never folds by an equality is* eventually vertical, *and is absorbed by vertical looping rather than by a horizontal crowd. (Exhaustive finite computation; machine-verified,* `wp38`*.)*

Proposition 1 and Lemma 2 explain why infinite *depth* is never the obstacle: forced structure is vertical, and vertical chains loop. Everything hard is horizontal—and, crucially, horizontal structure can be *exhibited* freely but seems impossible to *force* rigidly. A decision procedure must record only the horizontal branching it cannot reconstruct:

**Definition 3** (Shadows, live width, and F6)**.** In a model, or in a finite presentation of one, an edge is a *shadow* if its RCC5 value is forced by composition from edges already present (e.g. $x\,\mathsf{PP}\,y$, $y\,\mathsf{DR}\,z$ force $x\,\mathsf{DR}\,z$ by $\mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{DR}\}$); a shadow costs a decision procedure nothing. The *live width* at an interface is the number of non-shadow edges there—the genuine branching that must be recorded. Property **F6** states: for every satisfiable $C_0$, the live width of some presentation of a model of $C_0$ is bounded by a computable function of $C_0$.

# 4 History of $\mathcal{ALCI}_{\mathrm{RCC}}$ and why the known results do not settle it

## 4.1 Cohn's 1993 proposal

The idea of combining description logics with qualitative spatial reasoning via modal accessibility relations was first proposed by Cohn [1], who considered treating the RCC8 relations (or subsets thereof) as modalities in a multi-modal logic. The framework allowed spatial constraints to be expressed as modal formulas—e.g., $\langle\mathsf{PP}\rangle C$ for "some proper part satisfies $C$"—but Cohn left all decidability and complexity questions open.

## 4.2 Wessel's $\mathcal{ALCI}_{\mathrm{RCC}}$ family (2002/2003)

Wessel [7, 8, 5] gave the first rigorous treatment of the combination, introducing the $\mathcal{ALCI}_{\mathrm{RCC}}$ *family*: description logics $\mathcal{ALCI}_{\mathrm{RCCk}}$ ($k = 1, 2, 3, 5, 8$) extending $\mathcal{ALCI}$ with role boxes derived from the RCC composition tables. In an $\mathcal{ALCI}_{\mathrm{RCCk}}$ interpretation, every pair of domain elements is assigned exactly one RCC$k$ base relation, and the composition table is enforced globally (*strong* EQ *semantics*: EQ is identity). Models are complete graphs with RCC$k$ edge-colorings.

Wessel established the following:

- **Decidable cases:** $\mathcal{ALCI}_{\mathrm{RCC1}}$, $\mathcal{ALCI}_{\mathrm{RCC2}}$, and $\mathcal{ALCI}_{\mathrm{RCC3}}$ are decidable, because their composition tables are deterministic or near-deterministic, admitting reduction to standard DL techniques.
- **Lower bounds:** $\mathcal{ALCI}_{\mathrm{RCC5}}$ is PSpace-hard; $\mathcal{ALCI}_{\mathrm{RCC8}}$ is ExpTime-hard.
- **Open:** The decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ and $\mathcal{ALCI}_{\mathrm{RCC8}}$ was left as the central open problem.

In a series of companion reports, Wessel also mapped out the decidability landscape for description logics with general composition-based role inclusion axioms:

- $\mathcal{ALC}_{\mathcal{RA}}$ (arbitrary role axioms) is **undecidable**, via reduction from the Post Correspondence Problem [6].
- $\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$ (non-disjoint-roles variant) is also **undecidable**, via reduction from non-empty CFG intersection [3].

- $\mathcal{ALC}_{\mathcal{RA}_{SG}}$ (admissible role boxes: deterministic, functional, complete, and associative) is **decidable** (ExpTime-complete, via reduction to $\mathcal{ALC}$ with general TBoxes) [4]; adding unqualified number restrictions ($\mathcal{ALCN}_{\mathcal{RA}_{SG}}$) makes it undecidable again.

$\mathcal{ALCI}_{\mathrm{RCC5}}$ sits precisely in the gap: its role box is non-deterministic (unlike $\mathcal{ALC}_{\mathcal{RA}_{SG}}$) but has the patchwork property (unlike $\mathcal{ALC}_{\mathcal{RA}}/\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$).

**Remark 4** (Key difficulties)**.** $\mathcal{ALCI}_{\mathrm{RCC5}}$ has none of the properties that standard DL decidability proofs rely on: (i) no tree model property (models are complete graphs), (ii) no finite model property (some satisfiable concepts require infinite models), (iii) a non-deterministic role box (multi-valued composition entries).

For reference, the full RCC5 composition table is given in Table 1. Reading: row $= S(b,c)$, column $= R(a,b)$, entry = possible relations for $(a,c)$.

Table 1: The RCC5 composition table ($*$ = all five relations).

| comp | $\mathsf{DR}(a,b)$ | $\mathsf{PO}(a,b)$ | $\mathsf{EQ}(a,b)$ | $\mathsf{PPI}(a,b)$ | $\mathsf{PP}(a,b)$ |
|---|---|---|---|---|---|
| $\mathsf{DR}(b,c)$ | $*$ | DR, PO, PPI | DR | DR, PO, PPI | DR |
| $\mathsf{PO}(b,c)$ | DR, PO, PP | $*$ | PO | PO, PPI | DR, PO, PP |
| $\mathsf{EQ}(b,c)$ | DR | PO | EQ | PPI | PP |
| $\mathsf{PP}(b,c)$ | DR, PO, PP | PO, PP | PP | PO, EQ, PP, PPI | PP |
| $\mathsf{PPI}(b,c)$ | DR | DR, PO, PPI | PPI | PPI | $*$ |

Key composition facts exploited in the cover-tree theory (§6.3, §7): $\mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{DR},\mathsf{PPI}) = \{\mathsf{DR}\}$ (DR rigidity); $\mathsf{comp}(\mathsf{PP},\mathsf{PP}) = \{\mathsf{PP}\}$ and $\mathsf{comp}(\mathsf{PPI},\mathsf{PPI}) = \{\mathsf{PPI}\}$ (transitivity); and $\mathsf{comp}(\mathsf{DR},\mathsf{PO}) = \{\mathsf{DR},\mathsf{PO},\mathsf{PP}\}$ (PP can be *forced* by composition and universal filtering; see the forced-composition examples in §6.1).

### 4.3 Wessel's grid constructions and the coincidence obstruction

In his investigation of $\mathcal{ALCI}_{\mathrm{RCC8}}$ [8], Wessel made two important discoveries:

**1. Even-odd chains (Section 4.4.2).** The TPP/NTPP distinction in RCC8 allows the construction of *functional successor chains*: each node alternates between *even* and *odd*, and a node can distinguish its direct TPPI-successor from all indirect ones by parity. Wessel used this to encode a binary $n$-bit counter, proving $\mathcal{ALCI}_{\mathrm{RCC8}}$ is ExpTime-hard.

**2. The grid coincidence obstruction (Section 4.4.3, Figure 11).** Wessel constructed explicit RCC8 networks isomorphic to $n \times n$ grids, using TPP for horizontal and vertical successors. However, he identified a fundamental obstacle to encoding domino tilings: the *coincidence condition* $R_X \circ R_Y = R_Y \circ R_X$ (east-of-north equals north-of-east) cannot be enforced by concept terms. The composition table allows $\{PO, EQ, TPP, NTPP, TPPI, NTPPI\}$ at the coincidence point—too many options. Wessel concluded:

> "Even though these infinite models do exist, we have found no way to *enforce* the depicted model(s) by means of a concept term."

Wessel further observed that adding a hybrid logic binding operator $\downarrow$ (nominals) would immediately yield undecidability, confirming that the coincidence condition is the *precise boundary.*

Table 2: Why standard undecidability reductions fail for $\mathcal{ALCI}_{\text{RCC5}}$.

| **Source Problem** | **Technique** | **Missing in $\mathcal{ALCI}_{\text{RCC5}}$** |
|---|---|---|
| $\mathbb{N} \times \mathbb{N}$ domino | Graded modalities + transitivity | Number restrictions |
| $\mathcal{ALC}_{\mathcal{RA}}$ [6] | Post Correspondence Problem | Arbitrary role axioms |
| $\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$ [3] | Non-empty CFG intersection | Non-disjoint roles |
| $\mathcal{ALCI}_{\text{RCC8}}$ (Wessel) | Grid via $\mathsf{TPP}/\mathsf{NTPP}$ | $\mathsf{TPP}/\mathsf{NTPP}$ distinction |
| $L_{RCC8}$ (Lutz–Wolter) | Linearized topological enumeration | Topological rigidity |
| $L_{RCC5}(RS^{\exists})$ | $S5^3$ via supremum regions | Supremum closure |

Wessel thus independently developed, three to four years before Lutz–Wolter, the RCC8 description-logic analogues of these ingredients—the even-odd chain, an explicit two-dimensional $\mathsf{TPP}/\mathsf{NTPP}$ grid, and the first identification of the coincidence gap as the decisive obstacle. Lutz–Wolter later obtained undecidability by a *different* route—a linearized tiling enumeration (their §5, following Marx–Reynolds) that sidesteps the two-dimensional coincidence rather than enforcing it—so this is shared lineage and prior identification, not a dependence of their proof on Wessel's constructions.

### 4.4 Lutz & Wolter (2006)

Lutz and Wolter [9] studied *modal logics of topological relations*, denoted $L_{RCC8}$, $L_{RCC5}$, etc. These have the same *syntax* as $\mathcal{ALCI}_{\text{RCC}}$ but are interpreted over *topological models*: the domain elements are regular closed subsets of a topological space (typically $\mathbb{R}^n$). They proved:

- $L_{RCC8}$ is **undecidable** (their §5: a linearized $\mathsf{NTPP}$-chain enumeration of tile positions, following Marx–Reynolds, whose faithfulness relies on the geometry of $\mathbb{R}^n$; no two-dimensional grid coincidence is enforced).
- $L_{RCC5}(RS^{\exists})$ is **undecidable** (reduction from $S5^3$ via supremum regions).
- $L_{RCC5}(RS)$ is **open**. Lutz and Wolter explicitly wrote (p. 31): "Perhaps the most interesting candidate is $L_{RCC5}(RS)$ [...] to which the reduction exhibited in Section 8 does not apply."

The logic $\mathcal{ALCI}_{\text{RCC5}}$ is exactly $L_{RCC5}(RS)$: arbitrary complete-graph models with composition-table semantics. The problem has been recognized as open by leading researchers since 2006.

## 5 Why standard undecidability reductions fail

Undecidability proofs for description logics use a variety of techniques: grid encoding ($\mathbb{N} \times \mathbb{N}$ domino tiling, requiring functional roles or number restrictions), reduction from the Post Correspondence Problem (PCP), reduction from context-free grammar (CFG) intersection, or simulation of $S5^3$ via geometric constructions. $\mathcal{ALCI}_{\text{RCC5}}$ lacks the features that each of these techniques exploits: it has no functional roles, no number restrictions, no arbitrary role box, and no geometric coincidence conditions.

### 5.1 Blocked reduction routes

Table 2 summarizes the blocked routes. The **patchwork property** of RCC5 (Renz & Nebel [2])—local (triple-wise) consistency implies global consistency for atomic networks—actively resists grid encoding, since tiling reductions require rigid global constraints that go beyond local consistency.

### 5.2 A concrete domino attempt in $\mathcal{ALCI}_{\text{RCC5}}$

Can we encode a domino-like grid using $\mathsf{PP}/\mathsf{PPI}$ chains in RCC5? Consider the concept:

$$\begin{aligned} X \;=\; & \exists\mathsf{PPI}.\exists\mathsf{PPI}.C \;\sqcap\; \exists\mathsf{PPI}.\exists\mathsf{PPI}.D \\ & \sqcap\; \forall\mathsf{PPI}.\forall\mathsf{PPI}.(\forall\mathsf{PO}.(\neg C \sqcap \neg D) \sqcap \forall\mathsf{DR}.(\neg C \sqcap \neg D) \\ & \qquad\qquad\qquad \sqcap\; \forall\mathsf{PPI}.(\neg C \sqcap \neg D) \sqcap \forall\mathsf{PP}.(\neg C \sqcap \neg D)) \\ & \sqcap\; \forall\mathsf{PPI}.\exists\mathsf{PPI}.X \end{aligned}$$

The intent: each $X$-node has two $\mathsf{PPI}$-grandchildren colored $C$ and $D$ (the "square"), and the universals isolate the colored nodes from all non-$\mathsf{EQ}$ neighbors. The recursive $\forall\mathsf{PPI}.\exists\mathsf{PPI}.X$ forces infinite repetition.

**One level is satisfiable.** Without the recursive conjunct, $X$ has a 3-element model: $z \;\mathsf{PP}\; y \;\mathsf{PP}\; x$, with $z \in C \sqcap D$. The two grandchildren are forced to be *the same element* ($\mathsf{EQ}$ collapse): if $z_1 \neq z_2$, they would be related by some $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, and the universals would require $z_2 \notin D$ (firing $z_1$'s $\forall R.\neg C \sqcap \neg D$). So $z_1 = z_2 \in C \sqcap D$, and the universals are vacuously satisfied (the only neighbors of $z$ are PP-predecessors, not PO/DR/PPI-neighbors).

**Two levels is unsatisfiable.** Adding $\forall\mathsf{PPI}.X$: the $\mathsf{PPI}$-child $y$ must itself satisfy $X$ (without recursion), producing a new $C \sqcap D$-labeled element $w$ as a $\mathsf{PPI}$-child of $z$. But $z$ is a $\mathsf{PPI}$-grandchild of $x$, and $x$'s universals fire: $z$'s $\mathsf{PPI}$-neighbor $w$ must satisfy $\neg C \sqcap \neg D$. Yet $w \in C \sqcap D$. **Contradiction.**

This was verified computationally: a simplified version $X_1 = \exists\mathsf{PPI}.\exists\mathsf{PPI}.C \sqcap \forall\mathsf{PPI}^3.\neg C$ is SAT; $X_2 = X_1 \sqcap \forall\mathsf{PPI}.X_1$ is UNSAT. The domino encoding collapses at depth 2 because the isolation constraints from the grandparent level kill the colored elements created by the recursive unfolding.

**Remark 5.** The failure is not accidental. In $\mathcal{ALCI}_{\text{RCC5}}$, the only way to create "functional" successors is via $\mathsf{PP}/\mathsf{PPI}$ chains, but $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ makes these chains *transitive*—grandparent universals automatically propagate to grandchildren, preventing the positional diversity needed for grid encoding. In $\mathcal{ALCI}_{\text{RCC8}}$, the $\mathsf{TPP}/\mathsf{NTPP}$ distinction breaks this transitivity (a direct $\mathsf{TPPI}$-child is distinguishable from indirect descendants), which is why Wessel's even-odd chain works there but has no RCC5 analogue.

### 5.3 A concrete domino attempt in $\mathcal{ALCI}_{\text{RCC8}}$

The remark above suggests the $\mathsf{TPPI}/\mathsf{NTPPI}$ distinction in $\mathcal{ALCI}_{\text{RCC8}}$ might rescue the construction: a direct successor ($\mathsf{TPPI}$) and an indirect one ($\mathsf{NTPPI}$) are genuinely distinguishable, so perhaps the level-2 collapse can be avoided. We attempt the analogue and show that while it escapes the RCC5 failure mode, it fails for a different reason—Wessel's *coincidence obstruction* [8].

**The schema.** Let $\mathsf{Corner}$ be the concept that isolates a node from all non-$\mathsf{EQ}$ neighbors:

$$\mathsf{Corner} \;=\; \bigwedge_{R \in \{\mathsf{DC},\mathsf{EC},\mathsf{PO},\mathsf{TPP},\mathsf{NTPP},\mathsf{TPPI},\mathsf{NTPPI}\}} \forall R.(\neg C \sqcap \neg D).$$

The isolation is placed *locally on the colored grandchildren* rather than as an ancestral $\forall\mathsf{TPPI}.\forall\mathsf{TPPI}$ universal, so it cannot propagate transitively through the recursion:

$$\begin{aligned} X \;=\; & \exists\mathsf{TPPI}.\exists\mathsf{TPPI}.(C \sqcap \mathsf{Corner}) \;\sqcap\; \exists\mathsf{TPPI}.\exists\mathsf{TPPI}.(D \sqcap \mathsf{Corner}) \\ & \sqcap\; \forall\mathsf{TPPI}.(\neg C \sqcap \neg D) \;\sqcap\; \forall\mathsf{TPPI}.X. \end{aligned}$$

The third conjunct $\forall\mathsf{TPPI}.(\neg C \sqcap \neg D)$ forces the corner relation $x \to z$ to be $\mathsf{NTPPI}$ (not $\mathsf{TPPI}$), using $\mathsf{comp}(\mathsf{TPPI}, \mathsf{TPPI}) = \{\mathsf{TPPI}, \mathsf{NTPPI}\}$.

**Level 1 is satisfiable, with a real direct/indirect distinction.** Without recursion, $x$ has $\mathsf{TPPI}$-children $y_1, y_2$ (the two "direct" positions) and a single corner $z$ reached by $\mathsf{TPPI}$-$\mathsf{TPPI}$ chains from both $y_1$ and $y_2$. The EQ-collapse at level 2 works exactly as in RCC5: $z_1 \in C \sqcap \mathsf{Corner}$ and $z_2 \in D \sqcap \mathsf{Corner}$ are mutual non-$\mathsf{EQ}$ neighbors only if $\mathsf{Corner}$ is violated, so under strong-$\mathsf{EQ}$ semantics $z_1 = z_2 = z$. Crucially, $z \neq y_1, y_2$ in general: $z$ is $\mathsf{NTPPI}$ of $x$ while $y_1, y_2$ are $\mathsf{TPPI}$—the direct/indirect distinction survives, which is a genuine RCC8 refinement over the RCC5 attempt of §5.2.

**Level 2 collapses via the coincidence obstruction.** Adding $\forall\mathsf{TPPI}.X$: the $\mathsf{TPPI}$-child $y$ of $x$ satisfies $X$, producing its own corner $w \in C \sqcap D \sqcap \mathsf{Corner}$ at $\mathsf{NTPPI}$-distance from $y$, and hence at $\mathsf{NTPPI}$-distance from $x$ via $\mathsf{comp}(\mathsf{TPPI}, \mathsf{NTPPI}) = \{\mathsf{NTPPI}\}$. Now $x$'s own corner $z$ and $y$'s corner $w$ are distinct grid positions in the intended reading—but as elements of the model they are related by some RCC8 base relation. Since $z$ carries $\mathsf{Corner}$, every non-$\mathsf{EQ}$ neighbor of $z$ lies in $\neg C \sqcap \neg D$. But $w \in C \sqcap D$. Therefore $z = w$ under strong-$\mathsf{EQ}$. **All corners across the grid coincide at a single point.**

**Why the $\mathsf{TPPI}/\mathsf{NTPPI}$ trick fails.** In the RCC5 attempt the collapse is a *transitive-propagation* failure: the grandparent universal fires on great-grandchildren because $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$. Moving the isolation to a local $\mathsf{Corner}$ marker and using the non-transitive piece $\mathsf{comp}(\mathsf{TPPI}, \mathsf{TPPI}) = \{\mathsf{TPPI}, \mathsf{NTPPI}\}$ eliminates that specific pathway—the universal at the grandparent no longer reaches deeper nodes. But a *different* obstruction takes over: the composition table permits any of $\{\mathsf{PO}, \mathsf{EQ}, \mathsf{TPP}, \mathsf{NTPP}, \mathsf{TPPI}, \mathsf{NTPPI}\}$ between two corners at different intended grid coordinates, and nothing in the concept language rules out $\mathsf{EQ}$. The $\mathsf{Corner}$ markers themselves then force the EQ-merge. This is exactly the obstruction Wessel identified in report7.pdf, Figure 11 (cf. §4.3): the composition table does not carry the rigidity needed to keep distinct grid cells distinct.

**Remark 6.** The failure mode is structurally different from the RCC5 one but traces back to the same root cause. Lutz–Wolter's $L_{RCC8}$ reduction *circumvents* the coincidence obstruction rather than enforcing it: they linearize the plane into a single $\mathsf{NTPP}$-chain (the Marx–Reynolds enumeration) and recover the vertical direction from the enumeration, so no "north-of-east = east-of-north" coincidence has to be forced; the geometry of $\mathbb{R}^n$ only has to keep that linear encoding faithful [9]. No such rigidity is available in the abstract RCC8 semantics of $\mathcal{ALCI}_{\mathrm{RCC8}}$: a pure composition table allows corner points to merge freely, and the $\mathsf{TPPI}/\mathsf{NTPPI}$ distinction—while useful locally for a direct/indirect separation at level 2—cannot be chained to enforce an unbounded $\mathbb{N} \times \mathbb{N}$ grid. This matches the open-problem status of $\mathcal{ALCI}_{\mathrm{RCC8}}$ decidability noted in [9] and further discussed in [8].

## 6 Overview of approaches, by technique

We organize the attack by *technique*, in roughly the order tried, not by round or date. Each subsection follows the same shape: the high-level idea; the concrete techniques investigated; where and why they fail (with the examples that made the failure clear); pointers to the papers, probes, and reasoners; and the lessons.

### 6.1 Naive tableaux and blocking

**The idea, and why it should work.** The default decision method for a description logic is a *tableau*: to test whether $C_0$ is satisfiable, try to build a model top-down. Start with one individual required to satisfy $C_0$; whenever a node must satisfy $\exists R.D$, create a fresh $R$-successor satisfying $D$; whenever it satisfies $\forall R.D$, propagate $D$ to all its $R$-successors; close a branch on a contradiction. Left alone this may never terminate—$\exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top$ builds an infinite $\mathsf{PP}$-chain—so one adds *blocking*: if a new node's "type" (the set of concepts it must satisfy) repeats that of an ancestor, we stop expanding it and reuse the ancestor's already-built subtree, folding the infinite chain into a finite loop. For ordinary modal and description logics this is sound and complete, and it is the first thing one tries here. Our most refined implementation is the *cover-tree tableau* (§7), which blocks on *exact occurrences.*

**Why the total-relation setting breaks it.** In $\mathcal{ALCI}_{\mathrm{RCC5}}$ *every* pair of individuals carries a relation, so a model is a complete graph, not a tree. This single fact defeats ordinary blocking in three compounding ways, each of which we hit in practice.

1. **Composition forces edges *into* already-built nodes.** Creating a new node $z$ does not just attach it to its parent; by composition it forces relations between $z$ and *every* existing node. From $x\,\mathsf{PP}\,y$ and $y\,\mathsf{DR}\,z$ the table forces $x\,\mathsf{DR}\,z$ (Proposition 1). So the "tree" is a fiction: the real object is the whole complete graph, and a node's obligations depend on its relations to the entire context, not on its concept type alone.

2. **Dormant universals reawaken blocked nodes.** A restriction $\forall R.D$ lies inert at a node until an $R$-edge from it appears. But a *forced* composition edge (point 1) can create such an edge *after* the node was blocked, firing the dormant universal, adding $D$ to a node downstream, and possibly *unblocking* it. The cascade of re-qualifications need not terminate, and—worse—may be *unsound* if ignored: the block asserted a repetition that the later edge invalidates.

3. **Local consistency is not extensibility (the one-point extension problem).** Even a configuration in which every triangle is composition-closed may admit *no* valid fresh witness. A minimal instance: old nodes $a, b, c$ with $a\,\mathsf{DR}\,b$, $a\,\mathsf{PPI}\,c$, $b\,\mathsf{DR}\,c$, all consistent; requiring a new $d$ with $b\,\mathsf{PO}\,d$ and $c\,\mathsf{PO}\,d$ forces, through the two triangles, $a\,\mathsf{DR}\,d$ *and* $a\,\mathsf{PPI}\,d$ simultaneously—a contradiction invisible to any check that looks only at the existing network.

**The ladder of blocking conditions we tried.** Each rung tracks more relational context, and each is defeated by the phenomena above. The base choice is *where* one may block: the obvious strengthening of ancestor blocking is *anywhere blocking*—block a node against *any* earlier node in the tableau, not only its ancestors, the more aggressive termination device standard for logics with inverse roles. This helps termination but not soundness: it inherits every failure below, since it still compares nodes by a *local* summary while the constraints are global. On top of the where-question we layered progressively richer *what*-summaries. *Subset/type* blocking (repeat of the concept set) ignores relations entirely and is unsound at once. *Triangle* blocking tracks, for each node, the composition-*triangles* it participates in—the relation triples $(\rho(x, y), \rho(y, z), \rho(x, z))$—so that composition constraints are visible; but a triangle is a three-point view, and forced edges from a fourth point (via inverse roles) escape it (`triangle_blocking_ALCIRCC5`, `src/triangle_*.py`). *Quadruple* types lift this to four points to capture those cross-edges (`decidability_via_quadruples_ALCIRCC5`, `src/quadruple_type_check.py`); but the required arity is not bounded—a forced constraint can span a chain of any length. *Profile* blocking records a node's type together with the multiset of its relations to a bounded context (`src/profile_blocking_*.py`); but the context that actually determines future obligations is not bounded a priori—this is F6 in disguise. *Double* blocking (the $\mathcal{SHIQ}$ device: block only when a node *and its predecessor* jointly repeat an ancestor pair, which is what one needs for inverse roles) addresses point 2's predecessor sensitivity but not the forced-edge and extension problems (`src/double_blocking_test.py`).

**The examples that settle it.** Two families are diagnostic and recur throughout the project. The forced-composition UNSAT concepts, e.g. $C_{\text{force}} = \exists \mathsf{PP}.(\exists \mathsf{DR}.A) \sqcap \forall \mathsf{DR}.\neg A$: here $x\,\mathsf{PP}\,y$ and $y\,\mathsf{DR}\,z$ force $x\,\mathsf{DR}\,z$, so $\forall \mathsf{DR}.\neg A$ kills the required $z : A$; a procedure that ignores forced edges wrongly *accepts* it. Dually, the strong-$\mathsf{EQ}$ $\mathsf{PO}$-*loop* $C \sqcap \exists \mathsf{PO}.\exists \mathsf{PO}.C \sqcap \forall \{\mathsf{PO}, \mathsf{DR}, \mathsf{PP}, \mathsf{PPI}\}.\neg C$ is satisfiable *only* via a two-element model closed into a symmetric $\mathsf{PO}$-loop under $\mathsf{EQ}$-identity; a procedure that distinguishes occurrences by type alone wrongly *rejects* it (this is precisely the blind spot of the quasimodel oracle, §6.2). Further diagnostics—`inter_witness_counterexample_ALCIRCC5`, `typed_patchwork_counterexample_ALCIRCC5`—exercise the same seams.

**Lesson.** In a total-relation logic, sound termination cannot be had by *local* blocking, because the relational completion is *global*: what may be reused depends on the whole graph, and forced edges plus dormant universals make "the whole graph" irreducible to any fixed-arity local view. The way out is to stop trying to block a complete graph and instead *separate* the tree skeleton from the global relational completion, controlling the latter by the RCC5 patchwork property. That is the split-forest idea (§6.3).

### 6.2 Finite-model and reduction attempts

**The idea.** If one could force models to be finite, or finitely presented, one could enumerate candidates and check them. Four concrete lines: build a *quasimodel* (a set of types with a consistency relation) and eliminate types that cannot be realized; collapse infinite $\mathsf{PP}/\mathsf{PPI}$ chains into finite *clusters or loops*; take a *two-tier quotient* (one "kernel" per recurrent chain descriptor, plus finitely many designated witnesses); or *encode* satisfiability into monadic second-order logic (MSO) over trees and invoke Rabin's theorem.

**Where they fail.** The quasimodel / type-elimination procedure (`decidability_ALCIRCC5`, `src/alcircc5_reasoner.py`) is *retracted*: its elimination rule for $\mathsf{PO}$-demands ($Q3$) is *anti-monotone*—eliminating one type can invalidate the witness that justified another—so the fixpoint over-eliminates and the procedure is *incomplete* (though not unsound). Concretely it is wrong on the $\mathsf{PO}$-loop above: it reports UNSAT for a satisfiable concept. The finite-cluster and MSO routes both stumble on the same rock: collapsing a $\mathsf{PP}$-chain is fine (Lemma 2), but the *horizontal* width across clusters is not bounded, and MSO cannot define an unbounded horizontal matching—so a correct encoding again presupposes a width bound. Every "make it finite" variant reduces to producing a computable live-width bound, i.e. to F6 (Definition 3).

**The win.** One fragment escapes unconditionally. In the two-tier quotient, $\mathsf{PO}$ is the *only* relation whose recurrence along a chain needs an "all-phase" safety condition—because, unlike $\mathsf{DR}/\mathsf{PP}/\mathsf{PPI}$, a $\mathsf{PO}$-witness at one chain position need not remain a $\mathsf{PO}$-witness at another (the compositions $\mathsf{comp}(\mathsf{PP}, \mathsf{PO})$ and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PO})$ are not singletons). If the concept contains *no* $\forall \mathsf{PO}$ subformula, that safety condition is vacuous, the quotient is finite, and the width is bounded by a computable function of the concept.

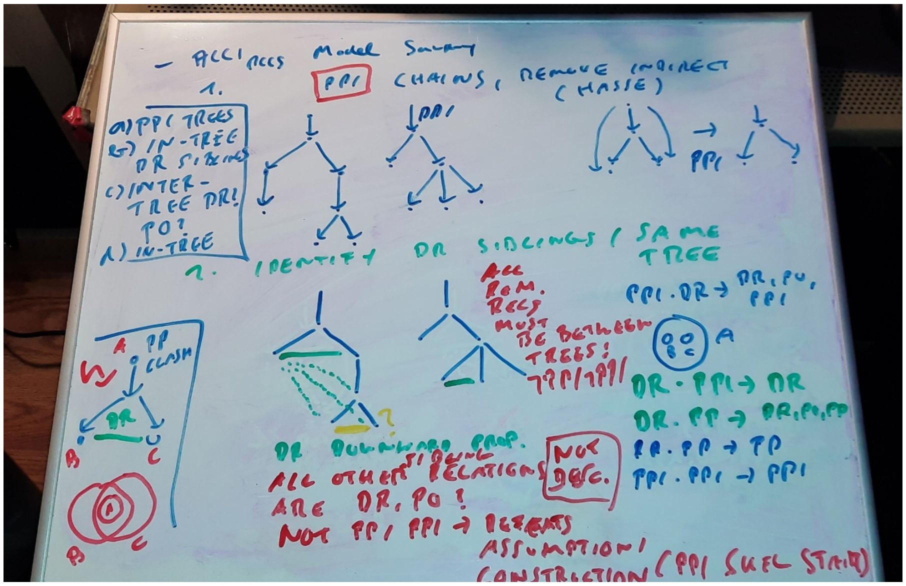


Figure 1: The original whiteboard sketch of the split-forest model decomposition (M. Wessel, April 2026): the intuition that a complete-graph RCC5 model can be presented as a forest of trees by splitting join nodes into $\mathsf{EQ}$-mates and completing the relational graph by patchwork. This idea—the human author's—is the semantic foundation that every subsequent proof route consumes, and the one component no cold review ever broke.

**Theorem 7** (Decidable fragment)**.** *Concept satisfiability for the* $\forall\mathsf{PO}$*-free fragment of* $\mathcal{ALCI}_{\mathrm{RCC5}}$ *—concepts with no* $\forall\mathsf{PO}$ *subformula, but retaining* $\exists\mathsf{PO}$*,* $\forall\mathsf{DR}$*,* $\exists\mathsf{DR}$*, and all vertical modalities—is decidable, with a computable live-width bound* $K(C_0) = (1+m)^2\, 2^{2^m}$*,* $m = |\mathsf{cl}(C_0)|$. (`two_tier_quotient_ALCIRCC5`, `po_free_fragment_ALCIRCC5`; theorem-level, resting on the two-tier quotient lemmas.)

The fragment is genuinely expressive—one may still say "overlaps some $A$" ($\exists\mathsf{PO}.A$), "all disjoint things are $B$" ($\forall\mathsf{DR}.B$), and impose arbitrary part-of nesting—so this is the project's strongest unconditional $\mathcal{ALCI}_{\mathrm{RCC5}}$ result.

**Lesson.** Finiteness is the wrong target: models are legitimately infinite (unbounded $\mathsf{PP}$-towers are satisfiable). The right target is a finite *presentation* of a possibly-infinite model.

### 6.3 The split-forest idea

**Idea (the human author's, on a whiteboard).** The split-forest decomposition was the human author's contribution, sketched on a whiteboard (Figure 1) and handed to the AI assistants to make precise. The insight: even though a model is a complete graph, one can *present* it as a *forest of trees*—one tree per connected component of the vertical ($\mathsf{PP}$) order—by *splitting* each node that is a common part of several superparts into $\mathsf{EQ}$-mates, so the tree skeleton becomes explicit, and then *completing* the missing (horizontal) edges by the RCC5 patchwork property. Tree-like presentations of infinite models are the classical gateway to decidability: once a logic has a tree-model property, automata or type-elimination usually finish the job. The bet was that this setting, despite its complete relational graph, still admits such a presentation.

**The normal form (Theorem A).** Made precise, the construction is a chain of thinning and splitting operations that every satisfiable concept survives: *witness thinning* (keep one witness per demand) → *generated containment covers* (organize witnesses by the vertical order) → *split copies* for incomparable superparts → *side-interface mosaics* (the horizontal relations at each interface, completed by patchwork) → a *typed*-EQ *quotient* (re-identify the split mates by exact occurrence). The output is a witness-generated, tree-like split-forest model. This normal form is the one component found sound by *every* cold review across the project—the semantic foundation everything downstream *consumes* rather than replaces (`trees/`, `cover_tree_tableau_ALCIRCC5`).

**Why it does not, by itself, decide.** The split forest gives the *shape* of models; it does not give a terminating *search*. Turning "a tree-like model of this shape exists" into "an algorithm can decide whether one exists" requires a *finite abstraction* of the (still infinite) split forest—and that finite-abstraction problem is exactly where the next two families live, and where the difficulty concentrates.

## 6.4 The no-automata certificate, and the automata route

**The hand certificate (rounds 2–10).** The first attempt at a finite abstraction was a hand-built *certificate*: a finite object whose validity (checked by local conditions on converse, composition, universal safety, request discharge, and residual frontiers) was meant to be equivalent to satisfiability (`gpt5.5_round2/`, `gpt5.5_round4/`). Across ten rounds it worked only by importing, piece by piece, machinery that is really *automaton-shaped*: the conditions that keep a certificate finite while a model is infinite are "request-closed cycles" (a demand and its witness recur), which behave exactly like Büchi/parity acceptance, and "finite product-state exhaustion," which behaves like reachability in a product automaton. Cold reviews found a succession of gaps—an infinite ancestor PPI-tower the construction missed (D-1), then its dual descendant PP-tower, then boundary non-determinacy, then (twice) that "finite checkability" was asserted rather than proved. The honest reading is that the hand certificate was an automaton written informally; so the project made the automaton explicit.

**The parity-automaton route (rounds 11–12).** This organizes the argument into three theorems. **Theorem A**: the split-forest normal form (above). **Theorem B**: a *finite abstraction* (frontier/pair/triple summaries of the split forest) captures saturated split forests up to satisfiability of all closure formulas. **Theorem C**: a two-way alternating parity *tree* automaton accepts exactly the valid abstractions, and its emptiness is decidable (Vardi). The automaton *consumes* the split forest; it does not replace it. Theorem B is the single load-bearing keystone. Cold reviews (the 5th and 6th) found its global-coherence step *assumed, not proved*: it derived a globally consistent RCC5 labelling of an infinite tree from purely local checks, via a "confluence law" that was never verified. Round 12 rebuilt the abstraction on *patchwork bags* (finite complete atomic networks that agree on overlaps), so global consistency follows from the citable RCC5 *patchwork property* (Renz–Nebel), and the countable completion by a compactness argument over the finite relation alphabet; eventualities are decided by two-way emptiness. A 7th (hot) review then found a residual *Shadow Amalgamation* gap in the truth lemma: the declared shadow edges were asserted to be jointly realizable with the bag networks, which is false without a further discipline (`automata_route_repairs/`, `cold_review_theoremB/`, probes `wp13`–`wp17`).

**Why we abandoned the tree automaton.** A cold review of the decision layer exposed a structural problem. The automaton reads a labelled tree and decides accept or reject. But the tree does *not* carry all the data that makes an abstraction valid: it omits the relations between each fresh witness and the "sources" of the universal restrictions ($\forall R.D$) that act on it. A tree is valid exactly when *some* consistent choice of those missing relations exists—so the automaton would have to solve a *guessing* problem ("does some completion of the missing data work?"), not merely read off the tree. The trouble is how an alternating tree automaton works: it explores the tree in many independent parallel copies, and each copy would guess the missing relations *locally*, with nothing forcing different copies to guess *consistently*. One can therefore build a tree in which two copies, running over two different witnesses, each privately guess relations that are locally fine but mutually incompatible, so every copy accepts—even though the true relation between those two witnesses is forced by composition to be impossible ($\mathsf{comp}(\mathsf{PP},\mathsf{PP}) \cap \mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{PP}\} \cap \{\mathsf{DR}\} = \emptyset$). The automaton accepts an unsatisfiable input: acceptance without validity. The three natural repairs (annotate the tape—but the enrichment is unboundedly large; quotient the profiles—provably lossy; guess canonically—needs an unproven uniformization) are each blocked by facts the construction itself establishes. So the tree automaton was replaced by a mosaic / type-elimination procedure directly on the infinitary abstraction (§6.5). The keystone survived the change untouched: Theorem B's finite-abstraction adequacy is F6 in automaton clothing.

### 6.5 The Lean line and F6

**Lean, and how we use it.** Lean 4 is a proof assistant whose kernel type-checks every inference and refuses a proof with a gap. In a project where prose proofs repeatedly *assumed* the hard step, the kernel is a ground-truth oracle that cannot be talked past. We moved the load-bearing soundness argument into Lean 4 core (no external libraries); the development is `formal/Round19Transport.lean` (≈5,300 lines, zero `sorry`), with the full history in `LEAN.md`.

**The cluster-quasimodel decision layer (rounds 15–18).** Following the cold review that killed the tree automaton, the decision layer became a finite *catalogue* of interface patterns, each carrying a *steering function* that fixes the relations from old occurrences to fresh ones, with *one-step* demand fulfilment—every existential is satisfied by an actually-present witness, so there are no promissory obligations and hence no parity/eventuality condition at all. It was on this construction, finally explicit and free of hidden choices, that the entire remaining difficulty localized to two named lemmas: **F6** (Definition 3, a computable live-width bound) and **W2′** (that locally realizable glue steps factor through the finite catalogue's states).

**What is machine-checked (rounds 19–30).** The soundness pipeline is kernel-checked end to end: certificate syntax → an S-condition on the certificate → a proper atomic RCC5 frame → the logic layer (a truth lemma, with one-step fulfilment) → an actual RCC5 model of $C_0$. So is the *faithfulness* of the finite (Hintikka) abstraction: satisfiability and abstraction-realizability coincide. On top of these sits a fixed, non-oracular Boolean checker of first-order finite certificate data, proved sound, from which decidability follows by finite search—modulo exactly one premise, $\mathrm{F6} \wedge \mathrm{W2}'$.

**Theorem 8** (Conditional decidability)**.** *If F6 holds—the live width of models of every satisfiable $C_0$ is bounded by a computable function of $C_0$—then concept satisfiability for $\mathcal{ALCI}_{\mathrm{RCC5}}$ is decidable. The* soundness *half of this reduction (a valid finite certificate provably unfolds to a genuine RCC5 model of $C_0$) and the faithfulness of the Hintikka abstraction are* machine-checked in Lean 4 *(zero `sorry`).* (The uniformization W2′ folds into F6—see below—so the sole open premise is F6.)

**The keystone, sharpened.** Two cold attacks on F6 turned the vague "bounded width" into a precise dichotomy. The naive ways to force width—shell half-graphs, prefix pumps, single-chain

PO-gaps—all *collapse* to finite catalogues (they have $O(1)$ separator cover, a carried memory slot suffices). The sharp obstruction is an unbounded family of $M_n$ *identity-selector minors*: pairs $(L_i, U_i)$ each separated by a *private* selector $A_i$ and by no other, so the selector incidence matrix is the $n \times n$ identity—private matching of unbounded hitting number. F6 fails iff some concept *forces* such minors; F6 holds iff none does. And W2′, which looked like a second obstacle, was shown false in its coarse form but repairable by a state refinement whose size is bounded *by F6*—so it *folds into* F6, leaving a single open question (`fable5_width_barrier/`, `cold_review_f6_w2prime/`; probes `wp35`–`wp44`, incl. `wp44`: arbitrary RCC5 glue steps *can* have unbounded separator cover, so any F6 proof must use $\mathcal{ALCI}_{\mathrm{RCC5}}$-forceability, not RCC5-consistency alone).

### 6.6 The regular-cover pivot

**Idea.** A reframing suggested by the human author: a finite *quotient* of a PP-chain is invalid (a PP-cycle contradicts EQ-identity), but a finite *generator* is not—a state with a PP-self-loop *unfolds* to an infinite chain of distinct copies. So replace finite quotients by *regular covers*: finite presentations whose infinite *unfoldings* are the models, in the spirit of the two-tier quotient but for the full logic. This finer invariant can represent exactly the patterns that fool a width bound—identity selectors ($i = j$ via address equality), half-graphs ($i < j$ via an order test)—finitely.

**What it produced.** A complete *local* algebra of strong-EQ RCC5 networks, one piece of which is machine-*certified*:

**Theorem 9** (RCC5 normal form)**.** *A strong-*EQ *atomic RCC5 network is composition-closed if and only if it is an* ordered-disjoint structure*:* PP *is a strict partial order,* DR *is a symmetric, irreflexive, downward-closed disjointness (if $x\,\mathsf{DR}\,y$ and $x' \le x$, $y' \le y$ then $x'\,\mathsf{DR}\,y'$), comparable pairs are never disjoint, and* PO *is the residual. The forward direction (every RCC5 network is ordered-disjoint) is* machine-certified for arbitrary domains *(`formal/RCC5NormalForm.lean`, axioms `propext` only); the converse is exhaustively machine-checked up to size four (`wp47`).*

With it come free amalgamation of ordered-disjoint structures, an exact one-point extension criterion, and exact uniform/non-uniform cross-policy characterizations—together *exhausting* the local algebra (`rcc5_local_amalgamation_theory`, `regular_cover_route_pivot/`, probes `wp45`–`wp83`). The route independently re-derives Theorem 7.

**Why it is a better map, not a shorter path.** Its own keystone (a bounded coherent regular-cover property) is F6 relabelled, and the route concedes, after exhausting the local algebra, that the remainder "cannot be a purely local RCC5 composition problem." Because RCC5 relations are total, every model's Gaifman graph is complete, so a one-dimensional PP-chain already has unbounded clique number and treewidth; no off-the-shelf guarded or bounded-treewidth cover theorem applies, and the keystone would need a bespoke cover theorem for complete edge-coloured regular structures. The pivot is the right vocabulary for a future attack—but it does not move the summit.

## 7 Current implementation: reasoners

The project ships working reasoners, on the same foundation as the proof attempts but validated empirically rather than proved.

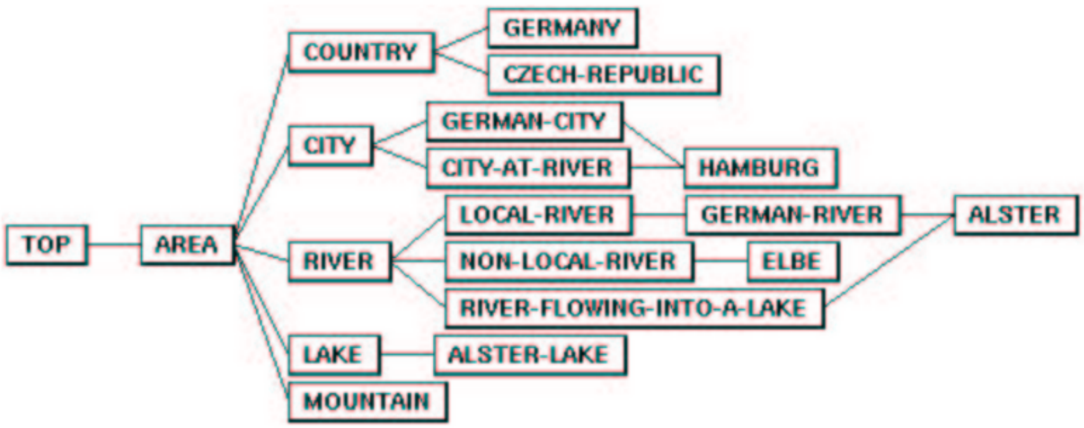


Figure 2: The GIS concept taxonomy of Wessel's 2002/2003 report (report7, Figure 6), recomputed in full (21/21 subsumptions) by the cover-tree tableau reasoner `src/gis_taxonomy.py`.

**The cover-tree tableau.** `src/cover_tree_tableau.py` ($\approx$ 350 lines) is an operational concept-satisfiability procedure with exact-occurrence blocking, resting on the split-forest normal form and patchwork completion (§6.3). It is cross-validated against an independent quasimodel oracle on 911 concepts (zero mismatches after the known blind spot is excluded) and against an independent model verifier (`src/model_verifier.py`). It is trustworthy on satisfiable inputs; its one known limitation is the strong-EQ PO-loop, decided correctly by the tableau but not by the quasimodel oracle. It is strong evidence, not a proof: no completeness theorem is claimed for the implementation.

**A GIS example.** As a sanity check spanning 23 years, the tableau recomputes the GIS concept taxonomy of Wessel's original report [8] (its Figure 6, reproduced here as Figure 2): all 21/21 subsumptions are reproduced (`src/gis_taxonomy.py`). A taxonomy computed by hand in 2003 is reproduced, in 2026, by an AI-written reasoner resting on the split-forest foundation.

## 8 On agentic AI usage

We report candidly on the workflow, because the methodology may outlast the mathematics.

**The loop.** The human posed the problem, supplied the central intuitions and directions (the split-forest decomposition, the cluster/loop and regular-cover reframings, the honesty discipline), and after each round reviewed, corrected, and re-prompted. An AI proposed a proof or a repair; a *fresh* AI session—with no memory of how the proof was built—then reviewed it adversarially. Repeat.

**Cold review as a standard step.** The single most transferable finding: a *cold* reviewer consistently found defects that a *warm* reviewer (primed by the construction) missed—including, on this project, unsoundness proofs and scope errors that two hot audits had passed over. Across fifteen reviews, a defect or overclaim was found in all but two. We recommend cold, cross-lineage adversarial review as a default intermediate step in AI-assisted mathematics, not an optional one.

**Don't trust, verify.** Every load-bearing claim was cross-checked against something mechanical: self-contained Python probes that re-derive the RCC5 composition table from finite set semantics (so the algebra cannot be fudged), and ultimately the Lean kernel. Overclaiming was treated as the cardinal sin—and it happened: the author model twice overstated what the Lean established, and a cold review caught both; the claims were withdrawn. The machine checks, not the model's confidence, are the ground truth.

**What frontier models did well, and did not.** Both models sustained long-context, rigorous, multi-round formal development and produced genuinely useful adversarial reviews and machine-checkable artifacts. What was *not* demonstrated is the decisive novel idea—no model cracked F6, and the honest expectation is that none will by the incremental means used here. The models excel at rigorous local formalization and at finding flaws; the genuinely new keystone is where human insight (or a qualitatively different search) is still required. We regard the AI contributions as co-authorship-grade in a human collaborator; we also regard the human's role—problem, intuition, direction, and the discipline of honest labelling—as indispensable.

# 9 Summary and conclusion

**What we learned.** A twenty-five-year-old open problem has been *reduced*, honestly and largely machine-checkably, to a single well-posed lemma (F6, bounded live width), which is provably the same fact that blocks a proof of undecidability. The local algebra of RCC5 is exhausted and, in one case, certified. Adversarial cold review plus machine verification is a repeatable methodology for this kind of work.

**What would move the needle.** F6 is a *definability* question—can a formula *force* unbounded rigid horizontal structure?—so the tools that fit are model-theoretic: locality (Gaifman/Hanf), Ehrenfeucht–Fraïssé / back-and-forth over bounded tuple profiles, the composition method, and bounded-model machinery. The concrete positive target is a *selector-sharing lemma*: that many same-profile selector gadgets must coalesce, so unbounded identity-selector minors cannot be forced. The concrete negative target is a concept forcing such minors (which would prove undecidability). Ordinary locality alone is insufficient because horizontal degree is unbounded; this is the "correct room," not a solved problem.

**The chances of closing F6.** We are frank: they are low, and re-treading the ground covered here will not change that. F6 is the open problem itself, and the two independent cold attacks converged on the same wall—evidence that the problem is correctly *mapped*, not that it is about to yield. It would be a poor use of effort to reconstruct the split-forest, automata, Lean, or regular-cover developments; the value of what remains is a genuinely new definability-theoretic idea, or a machine-assisted search of a different kind (below).

**The wins.** Independent of F6, this project produced: (i) an unconditionally *decidable fragment*, the $\forall\mathsf{PO}$-free concepts; (ii) a machine-*certified* RCC5 normal form; (iii) a machine-checked *conditional decidability theorem* (F6 $\Rightarrow$ decidable, soundness certified, resting on the F6 oracle); (iv) working *reasoners* reproducing a 2003 GIS taxonomy; (v) a sharp *map* of the problem with a named toolbox; and (vi) a candid *methodology* for adversarial AI-assisted mathematics.

**Is this a scientific contribution?** We believe so, with the labelling kept honest: not a resolution of the open problem, but its reduction—checkably, and with a certified surround—to an irreducible core, plus one unconditional fragment, one certified lemma, working tools, and an honest account of an AI-assisted attack and its dead ends. That is an accurate and, we think, respectable terminal state for a discussion piece on a problem open for over two decades.

**Outlook: the human is the bottleneck.** Although the key insights, intuitions, and directional choices came from the human, each round still required the human to orchestrate, review, and rewrite the prompts—and at that cadence roughly eighty rounds consumed about four months. Autonomous sub-agents *directing* the research could plausibly automate the diminishing-delta local work (formalization, local lemmas, probe-driven counterexample search) and remove that bottleneck. The cost may be prohibitive and it sacrifices direct oversight, but it is worth attempting for one reason this project is unusually well-suited to it: the objective is clearly defined and *machine-verifiable in Lean*. A closed, adversarial loop—propose, formalize, let the kernel accept or reject—has a built-in ground-truth oracle, exactly the guard the manual cold-review methodology provided. The honest caveat: Lean verifies *proofs*, not the *truth of a conjecture*, so such a loop can tighten the certified surround but the summit (F6) still needs a genuine idea; orchestration alone is unlikely to supply it.

**Materials.** The machine-checked development is `formal/Round19Transport.lean`, `formal/RCC5NormalForm.lean`, and `formal/ForcingReduction.lean` (see `LEAN.md`); the status report and the F6/W2′ analysis are in `papers/fable5_width_barrier/`; the decidable fragment in `papers/two_tier_quotient_ALCIRCC5` and `papers/po_free_fragment_ALCIRCC5`; the certified local theory in `papers/rcc5_local_amalgamation_theory`; the reasoners under `src/`; the probes under `verification/python/`; and the full audit trail in `CONVERSATION.md` and `OUTDATED.md`. The repository is at https://github.com/lambdamikel/alcircc5.

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## A On the DL 2026 rejection

An abstract of this work was submitted to the 2026 International Workshop on Description Logics and rejected. We record the episode not out of grievance alone but because it bears on the paper's own subject: what peer review is for, and how a community engages with an unusual result on one of its long-standing open problems.

The submission was labelled exactly as this paper is—*strongly supported, not certified*, an AI-assisted attack offered as a discussion piece. The reviews did not engage with the substance (the split-forest normal form, the reduction to F6, the machine-checked soundness), and no poster was granted. There is a recursive irony in an AI-assisted result on a twenty-year-old problem being declined without engagement in the same period that frontier AI began contributing checkable mathematics. The full open letter and the reviews are preserved in the repository (`dl2026-rejected/open_letter.md`, `reviews.txt`); we reproduce neither in full here, but stand by the position that a correctly-labelled, machine-checkable reduction of a two-decade open problem merits engagement on its merits.